

SURGERY FOR ISCHAEMIC HEART DISEASE (IHD)

Pathophysiology

- Progressive coronary blockage by **atherothrombotic disease**. Luminal compromise → supply/demand imbalance → **chronic stable angina**.
- **Plaque rupture + thrombosis** → **acute coronary syndromes (ACS)**: unstable angina, NSTEMI, STEMI.

Diagnosis

- Clinical, blood tests (troponin), **ECG**, treadmill, echo, **CT angiography**, **coronary angiography**.

Medical management

- **Aspirin**, **beta-blockers**, **nitrates**, **CCB** (if BB contraindicated), **statins** (plaque stabilization), **ACE inhibitors**, clopidogrel.

Acute management

- **STEMI**: **PCI (90-min door-to-balloon)** or thrombolytic (alternative). "**Time is myocardium**."
- **ACS**: aspirin + heparin; **clopidogrel** (better outcome if held 30 days... post-CABG); **prasugrel stopped 7 days pre-op**; **GP IIb/IIIa inhibitors (tirofiban, eptifibatide)** if continuing ischaemia, rest pain, HD instability, HF, ↑troponin, ECG changes.
- **PCI**: bare-metal stent (BMS) vs drug-eluting stent (DES).

Surgery (CABG)

- Surgery effective: relieves angina, delays infarction, **improves survival vs medical/PCI** in selected.
- **Indications — clinical**: class III/IV angina, ischaemic pulmonary oedema, failed PCI, patient undergoing major vascular surgery.
- **Indications — anatomic**: **left main >50%**, **3-vessel disease (3VD) with EF <50%**, 3VD EF >50% with inducible ischaemia, lesions not amenable to PCI.
- **Other**: >50% stenosis in branch arteries + valve surgery, MI complications, coronary anomalies with sudden death risk.
- **Conduits**: venous vs **arterial** (arterial superior — e.g. LIMA).
- **Procedures**: on-pump CABG, **OPCAB (off-pump)**, MIDCAB, TECAB, robotic, laser transmyocardial revascularization.
- **Post-op**: beta-blocker, aspirin, statin, ACE inhibitor.

MI complications (surgical)

- **Septal wall rupture**, **free wall rupture**, **ischaemic mitral regurgitation**, **LV aneurysm**.

High-yield one-liners

- **ACS** = plaque rupture + thrombosis (unstable angina/NSTEMI/STEMI).
- **STEMI** → **PCI 90-min door-to-balloon**; "**time is myocardium**".
- **CABG anatomic indications**: **left main >50%**, **3VD + EF <50%**, lesions not PCI-amenable.
- **Arterial conduit (LIMA)** superior to venous; **prasugrel stop 7 days pre-op**.
- **MI surgical complications**: **septal/free wall rupture**, **ischaemic MR**, **LV aneurysm**.
- **Post-op + medical IHD**: **aspirin + beta-blocker + statin + ACE inhibitor**.

SURGERY FOR ISCHAEMIC HEART DISEASE (Revision Table)

Bold = high-yield. Dr Shkar Saeed.

PATHOPHYSIOLOGY, DIAGNOSIS & MEDICAL

Topic	Key points
Pathophysiology	Atherothrombotic coronary blockage → stable angina ; plaque rupture + thrombosis → ACS (unstable angina/NSTEMI/STEMI)
Diagnosis	Clinical, troponin, ECG, treadmill, echo, CT angio , coronary angio
Medical	Aspirin , beta-blocker , nitrate , CCB , statin (plaque stabilization) , ACE inhibitor , clopidogrel
STEMI	PCI (90-min door-to-balloon) or thrombolytic; "time is myocardium"
ACS	Aspirin + heparin; clopidogrel; prasugrel stop 7 days pre-op ; GP IIb/IIIa (tirofiban) if ongoing ischaemia
PCI	BMS vs DES

SURGERY (CABG)

Topic	Key points
Benefit	Relieves angina, delays infarction, improves survival vs medical/PCI
Clinical indications	Class III/IV angina, ischaemic pulmonary oedema, failed PCI, major vascular surgery
Anatomic indications	Left main >50% ; 3VD + EF <50% ; 3VD EF>50% + inducible ischaemia ; lesions not PCI-amenable
Other	>50% branch stenosis + valve surgery, MI complications, coronary anomalies
Conduits	Venous vs arterial (LIMA superior)
Procedures	On-pump CABG, OPCAB (off-pump) , MIDCAB, TECAB, robotic, laser TMR
Post-op	Beta-blocker, aspirin, statin, ACE inhibitor
MI complications	Septal rupture , free wall rupture , ischaemic MR , LV aneurysm